Polynomial-Time Exact Learning of Fourier-Sparse Boolean Functions from Nearly Linear Quantum Samples

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Abstract

We give an algorithm that exactly recovers the Fourier representation of every Boolean function with at most k nonzero Fourier coefficients, using $O(k \log^3(2k))$ independent uniform quantum examples and polynomial total computation and working space in the input dimension and k. Each quantum example is measured separately. The algorithm uses no membership queries to the target and no inverse of a state-preparation circuit. Fresh residual labels estimate the coefficients used to weight a subsequent independent batch of Fourier samples. A majority of signs of the resulting averages is an explicitly evaluable Boolean hypothesis. Classical heavy-Fourier search on this known hypothesis produces candidate indices, which a separate label batch certifies against the original target. We prove a uniform contraction of missing original Fourier energy and account for adaptive refitting, exact rational arithmetic, failure probabilities and all resource costs. This complements the smaller sample bound of Ye by supplying a fixed polynomial computational guarantee. The bounds are asymptotic and do not assert practical running time.

1 Result, access model and contribution

Let $f : \mathbb{F}_2^n \to \{-1, 1\}$ and write

$$\chi_s(x) = (-1)^{s \cdot x}, \qquad c_s = \widehat{f}(s) = 2^{-n} \sum_x f(x)\chi_s(x), \qquad S = \operatorname{supp} \widehat{f}.$$

All norms and expectations on the cube use its uniform probability measure. The supplied quantum examples are independent copies of

$$|\psi_f\rangle = 2^{-n/2} \sum_x |x\rangle |b_f(x)\rangle, \qquad b_f(x) = (1 - f(x))/2. \tag{1}$$

The input consists only of these copies and integers n, k with $|S| \le k$. The cost of preparing the supplied examples is outside this access model; no preparation circuit or its inverse is supplied. A label batch means computational-basis measurements of fresh copies, giving independent uniform pairs $(X, f(X))$.

Theorem 1. *For $n \ge 1$ and $1 \le k \le 2^n$, every Boolean f with $|S| \le k$ can be learned exactly with probability at least $2/3$ using*

$$O(k \log^3(2k))$$

supplied uniform quantum examples. The output is the complete list of pairs (s, c_s) with $c_s \neq 0$, with exact rational coefficients. Total quantum work is $O(nk \log^3(2k))$. Conservative classical bit-time and live-space bounds are

$$\widetilde{O}(n^2k^7 + nk^9) \quad \textit{and} \quad \widetilde{O}(nk^2 + k^3), \tag{2}$$

respectively. Here only fixed powers of logarithms in n and k are suppressed. These resource caps hold on every measurement and random tape; an unsuccessful execution may return FAIL*. No chosen-input query to f is used.*

The theorem is about exact representation recovery, including coefficients that are rarely observed by Fourier sampling. It imposes no degree, Fourier-flatness or variance promise. The case $n = 0$ needs one label; an upper bound $k > 2^n$ can be replaced by 2^n.

Prior work and comparison. Arunachalam, Chakraborty, Lee, Paraashar and de Wolf [1] obtained an $O(k^{3/2} \log^2 k)$ quantum sample bound and a lower bound of $\Omega(k \log k)$ in their stated parameter range. Ye [2] improves the upper bound to $O(k \log^2 k / \log\log k)$ by constructing a grouped-sample candidate dictionary and proving support containment. That sample bound is smaller than ours. Ye's Theorem 1.1 states a sample guarantee; it does not state a fixed polynomial computation bound. Its Algorithms 1–2 explicitly construct and solve over the candidate dictionary. We do not assert a computational lower bound for that method. Caro et al. [3, Corollary 10] give an efficient exact learner with $O(k^4 \log(k^2/\delta))$ copies and success $1 - \delta$. Thus neither polynomial-time learning alone nor near-linear sample complexity alone is the contribution claimed here.

Our contribution is their simultaneous attainment through a computed approximation with error proportional to the current missing energy. We use Fourier sampling, Boolean granularity [4], concentration inequalities [6], and classical Fourier search [5] as attributed ingredients. The Fourier search is applied to a stored hypothesis, for which chosen-input evaluations are available. The target itself remains accessible only through supplied examples. We make no claim of optimal logarithmic factors or exhaustive publication priority.

Mechanism. The maintained approximation a has a certified support contained in S. Fresh labels estimate c_s at an error scale proportional to $\|f - a\|_2$. These estimates are fixed before drawing new Fourier samples. Thresholded reciprocals of the estimates then define an evaluable, bounded sign-majority hypothesis H. Searching H supplies candidate frequencies, including frequencies not directly sampled. Original-target certification, followed by exact refitting, contracts the missing energy. Keeping $f - a$, rather than $f - H$, as the residual is essential: H can have a dense Fourier spectrum.

2 Basic access and certificate lemmas

Parseval gives $\sum_s c_s^2 = 1$. We use the following consequence of Boolean granularity [4, Theorem 8.1]:

$$c_s \neq 0 \quad \Longrightarrow \quad |c_s| \geq 1/k. \tag{3}$$

For actual sparsity $s_0 > 1$, Theorem 8.1 of the cited April 27, 2011 author manuscript makes every coefficient an integer multiple of $2^{1-\lfloor \log_2 s_0 \rfloor}$. A nonzero coefficient therefore has magnitude at least $2/s_0 \geq 1/k$, proving (3). A sparsity-one Boolean function is a signed parity.

Apply Hadamard gates to all registers in (1). The amplitude of outcome $(s, 1)$ is $c_s/\sqrt{2}$. Thus the last bit is one with probability $1/2$, and conditional on that flag, s has probability c_s^2. Independent copies give independent flags and conditional samples. In particular, conditioning on enough flags in a fixed batch does not bias the successful indices.

For an archive $U \subseteq S$, put

$$W(U) = \sum_{s \in S \setminus U} c_s^2, \qquad P_U f = \sum_{s \in U} c_s \chi_s, \qquad \alpha^2 = 82/81. \tag{4}$$

The algorithm does not compute $W(U)$ or $P_U f$.

Lemma 2 (Fresh residual estimates)**.** *Let a be fixed before a fresh label batch, supported on at most k frequencies, and put $r = f - a$, $e = \|r\|_2$. Let $\mathcal{D}$ be any set of at most P frequencies fixed before that batch. For $0 < \varepsilon \le 1$, $0 < \beta < 1$, take*

$$q \ge 32k\varepsilon^{-2} \ln \frac{4(P+1)}{\beta}. \tag{5}$$

Define $v = q^{-1} \sum_\ell r(X_\ell)^2$ and $z_s = q^{-1} \sum_\ell r(X_\ell)\chi_s(X_\ell)$. With probability at least $1 - \beta$,

$$e^2/2 \le v \le 3e^2/2, \qquad |z_s - \widehat{r}(s)| \le \varepsilon e/\sqrt{k} \quad (s \in \mathcal{D}). \tag{6}$$

This conclusion also holds conditional on any history fixing $a, \mathcal{D}$ before the batch.

Proof. The residual has at most $2k$ Fourier terms, so $\|r\|_\infty \le \sqrt{2k}e \le \sqrt{3k}e$. Each $r(X)\chi_s(X)$ has variance at most e^2. Since $|\widehat{r}(s)| \le e$, its centered absolute bound is at most $(\sqrt{2k}+1)e \le 3\sqrt{k}e$. For the scalar bounded-case Bernstein inequality [6, Theorem 6.1(ii)], at deviation $t = \varepsilon e/\sqrt{k}$ the denominator satisfies

$$\sigma^2 + Bt/3 \le e^2(1+\varepsilon) \le 2e^2.$$

For q independent centered scalar variables bounded in absolute value by B, each with variance at most σ^2, that inequality bounds the deviation of their average by $2\exp[-qt^2/(2(\sigma^2 + Bt/3))]$. Thus a two-sided coefficient failure is at most $2\exp(-q\varepsilon^2/(4k))$. Also $\mathrm{Var}(r(X)^2) \le 3ke^4$ and $|r(X)^2 - e^2| \le 3ke^2$. At deviation $t = e^2/2$, the denominator is at most $2(3ke^4 + ke^4/2) = 7ke^4$, giving failure at most $2\exp(-q/(28k))$. Put $L_0 = \ln(4(P+1)/\beta)$. Equation (5) gives $q\varepsilon^2/(4k) \ge 8L_0$ and $q/(28k) \ge L_0$, so the union bound is at most $2(P+1)e^{-L_0} = \beta/2$. If $e = 0$, all residual values vanish and the conclusion is exact. □

Lemma 3 (Original-coefficient certification)**.** *Suppose $U \subseteq S$, a is supported on U, and a candidate list L, $|L| \le P$, precedes a fresh batch satisfying* (5)*. Admit $s \in L \setminus U$ precisely when*

$$z_s^2 > 4\varepsilon^2 v/k. \tag{7}$$

On (6)*, no frequency outside S is admitted, and every proposed frequency with $|c_s| > 4\varepsilon e/\sqrt{k}$ is admitted.*

Proof. Outside U the mean of z_s is c_s. For $c_s = 0$, its squared estimate is at most $\varepsilon^2 e^2/k$, while the threshold is at least $2\varepsilon^2 e^2/k$. If $|c_s| > 4\varepsilon e/\sqrt{k}$, the estimated magnitude exceeds $3\varepsilon e/\sqrt{k}$, and the square root of the threshold is at most $\sqrt{6}\varepsilon e/\sqrt{k}$. The zero-residual case admits nothing. □

Lemma 4 (One Gram event controls adaptive refitting)**.** *For a fresh batch of*

$$q \ge 300(k+1) \ln \frac{2(k+1)}{\beta} \tag{8}$$

uniform labels, with probability at least $1-\beta$, every subset $U \subseteq S$ has a nonsingular empirical Gram matrix, and its exact least-squares fit a satisfies

$$\|P_U f - a\|_2 \le \frac{\sqrt{W(U)}}{9}, \qquad \|f - a\|_2^2 \le \alpha^2 W(U), \qquad \|a\|_2^2 \le \alpha^2 < 2. \tag{9}$$

The set U may be chosen using this same batch. The empty fit is zero.

Proof. Let $S_0 = S \cup \{0\}$ and $p = |S_0| \le k+1$. The analysis-only row $\phi(X) = (\chi_s(X))_{s\in S_0}$ satisfies $\|\phi(X)\|_2^2 = p$ and $\mathbb{E}\phi(X)\phi(X)^{\mathsf{T}} = I_p$. The independent positive semidefinite summands $\phi(X_\ell)\phi(X_\ell)^{\mathsf{T}}$ have largest eigenvalue p and their expected sum is qI_p. Matrix Chernoff [6, Corollary 5.2] therefore bounds the lower and upper eigenvalue tails by

$$p\left[\frac{e^{-\delta}}{(1-\delta)^{1-\delta}}\right]^{q/p}, \qquad p\left[\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right]^{q/p}, \quad 0<\delta<1,$$

respectively. The scalar exponents satisfy

$$\delta + (1-\delta)\ln(1-\delta) \ge \delta^2/2, \qquad (1+\delta)\ln(1+\delta) - \delta \ge \delta^2/3.$$

The first follows by integrating $-\ln(1-t) \ge t$; the second by integrating $\ln(1+t) \ge t - t^2/2$ on $[0,\delta]$. Summing the tails gives

$$\Pr\left[\left\|q^{-1}A_{S_0}^{\mathsf{T}}A_{S_0} - I_p\right\|_{\mathrm{op}} > \delta\right] \le 2p\exp\left(-\frac{q\delta^2}{3p}\right), \quad 0<\delta<1.$$

Take $\delta = 1/10$; the budget in (8) is at least $300p\ln(2p/\beta)$ because both positive factors increase with p. This single event controls all principal submatrices and cross blocks, without a union bound over subsets. For $V = S \setminus U$, the normal equations give

$$a_U - c_U = (A_U^{\mathsf{T}}A_U/q)^{-1}(A_U^{\mathsf{T}}A_V/q)c_V.$$

The two operator norms are at most $10/9$ and $1/10$, respectively. This proves the first inequality; orthogonality proves the second. Finally

$$\|a\|_2 \le \sqrt{1-W(U)} + \frac{\sqrt{W(U)}}{9} \le \sqrt{1+1/81}.$$

The proof uses unknown S_0 only to define an event. The algorithm constructs only the columns it actually retains. If $U = S$, the fit is exactly f. □

3 Fourier search on a known Boolean function

We record the exact classical search interface needed below. It is the standard prefix-energy mechanism associated with Kushilevitz–Mansour [5]; the proof specifies the access and finite caps.

Lemma 5 (Capped heavy-spectrum search)**.** *Given deterministic evaluation access to $H : \mathbb{F}_2^n \to \{-1,1\}$, a rational threshold $0 < \tau \le 1$, and $0 < \beta < 1$, let $P = \lceil 4/\tau^2 \rceil$. A classical algorithm returns at most P indices, or* FAIL*, using at most $2nP$ prefix tests and*

$$N = \left\lceil 32\tau^{-4}\ln\frac{4nP}{\beta}\right\rceil$$

samples per test. Each sample needs two evaluations of H and at most $2n$ random bits. With probability at least $1-\beta$, it does not fail and its list contains every s with $|\widehat{H}(s)| \ge \tau$.

Proof. For a j-bit prefix u, orthogonality gives the bucket identity

$$\begin{aligned} E_u &= \sum_{v\in\mathbb{F}_2^{n-j}} \widehat{H}(u,v)^2 \\ &= \mathbb{E}_z\left(\mathbb{E}_x H(x,z)\chi_u(x)\right)^2 = \mathbb{E}_{z,x,y} H(x,z)H(y,z)\chi_u(x+y), \end{aligned} \tag{10}$$

where x, y are independent uniform j-bit strings, and the two evaluations share a uniform suffix z. This requires chosen inputs to H.

Start with the empty prefix. Test both children of each retained prefix with fresh samples of the last expression in (10); retain a child if its estimated energy is at least $3\tau^2/4$. Return FAIL immediately if the retained count exceeds P at any depth. Stop at depth n.

Each summand is in $[-1, 1]$, a range of length two. For one test, Hoeffding's average-deviation bound $2\exp(-Nt^2/2)$ at $t = \tau^2/4$ gives

$$\Pr[|\widetilde{E}_u - E_u| > \tau^2/4] \le 2\exp(-N\tau^4/32).$$

Condition on the past before each test and take a union bound over the tests. On simultaneous accuracy, every ancestor of a coefficient of magnitude at least τ survives. Every retained bucket has energy at least $\tau^2/2$; disjoint buckets at a fixed depth have total energy one. Hence at most $2/\tau^2 \le P$ survive and the cap does not fire. Fresh random inputs are generated classically and require no target examples. □

4 Constructing a queryable majority hypothesis

Assume $U \subseteq S$ and a fixed, known rational fit a satisfies (9). Write $W = W(U)$ and $e = \|f - a\|_2$. Draw an estimator label batch with Lemma 2, using list-size argument k in that lemma. For any index s, define

$$\widetilde{c}_s = \widehat{a}(s) + q_e^{-1} \sum_{\ell} (f(X_\ell) - a(X_\ell))\chi_s(X_\ell). \tag{11}$$

The formula is evaluated only at indices subsequently obtained from fresh Fourier measurements. Its values at all other indices need never be computed.

Conditional on the entire earlier history, S is fixed. Apply Lemma 2 to $\mathcal{D} = S$ in the analysis. This yields the event

$$e^2/2 \le v \le 3e^2/2, \qquad |\widetilde{c}_s - c_s| \le \Delta := \varepsilon e/\sqrt{k} \quad (s \in S), \tag{12}$$

with probability at least $1 - \beta$. The union bound does not require knowing S. It is not a claim of uniform estimation over all 2^n indices.

After fixing the estimator batch, draw g independent groups of m Fourier samples s_{ij}, retaining multiplicities. A frequency passes the cutoff if $\widetilde{c}_s \ne 0$ and

$$\widetilde{c}_s^2 \ge 16\varepsilon^2 v/k. \tag{13}$$

Define $w_s = 1/\widetilde{c}_s$ on a pass and $w_s = 0$ otherwise. Form

$$Q_i(x) = \frac{1}{m} \sum_{j=1}^{m} w_{s_{ij}} \chi_{s_{ij}}(x), \qquad H(x) = \text{majority}_{1 \le i \le g} \operatorname{sign} Q_i(x), \tag{14}$$

where g is odd and $\operatorname{sign}(0) = 1$. Zero estimates are never divided. This is an explicit finite classical circuit with exact rational weights. It has no call to f.

Lemma 6 (Bias and variance of the computed averages)**.** *Suppose $W > 0$ and (12) holds. Conditional on the estimator batch, let $h = \mathbb{E}Q_i$ as a function on the cube. Then*

$$\|h - f\|_2^2 \le 36\varepsilon^2 e^2, \qquad \operatorname{Var}(Q_i(x)) \le \frac{9k}{4m} \quad \textit{for every } x. \tag{15}$$

Proof. The cutoff magnitude $t = 4\varepsilon\sqrt{v/k}$ obeys $2\sqrt{2}\Delta \le t \le 2\sqrt{6}\Delta < 5\Delta$. On a pass, $|c_s/\widetilde{c}_s| \le 1 + \Delta/t < 3/2$ and the signs agree. On a failure, $|c_s| < 6\Delta$, including when the estimate is zero. Since future Fourier samples have probabilities c_s^2,

$$h(x) = \sum_{s \in S:\, s \text{ passes}} \frac{c_s^2}{\widetilde{c}_s} \chi_s(x).$$

The coefficient error on a pass is $c_s(c_s - \widetilde{c}_s)/\widetilde{c}_s$, of magnitude at most $3\Delta/2$. On a failure it is at most 6Δ. The passing and failing sets partition S, so Parseval bounds the total squared error by $36|S|\Delta^2 \le 36\varepsilon^2 e^2$. A single weighted sample has second moment $\sum_{s \text{ passes}} c_s^2/\widetilde{c}_s^2 \le 9k/4$, independently of x. Independence within each group proves the second inequality. The function h is analysis-only; the algorithm evaluates H, not h. □

Lemma 7 (Bounded majority and original residual scale)**.** *Take $m = 144k$. Conditional on* (12)*,*

$$\mathbb{E}_{\text{groups}} \|H - f\|_2^2 \le 576\varepsilon^2 e^2 + 4\, 2^{-g}. \tag{16}$$

If $\varepsilon^2 \le \beta/262144$ and $2^g \ge 512k^2/\beta$, then for $W > 0$,

$$\Pr_{\text{groups}}\Big[\|H - f\|_2^2 > W/64\Big] \le \beta. \tag{17}$$

Proof. Let $A = \{x : |h(x) - f(x)| > 1/2\}$. Its measure is at most $4\|h - f\|_2^2$. Outside A, h has the correct sign with margin at least $1/2$. Lemma 6 and Chebyshev bound each group's sign-error probability there by $1/16$. For a fixed such x, the groups are independent conditional on the estimator batch. Summing over possible erroneous majorities bounds their probability by $2^g(1/16)^{g/2} = 2^{-g}$. No independence over different inputs x is asserted. Integrating, and using the squared error bound four for Boolean outputs, gives

$$\mathbb{E}_{\text{groups}} \|H - f\|_2^2 \le 16\, \|h - f\|_2^2 + 4\, 2^{-g},$$

which proves (16). Since $e^2 \le \alpha^2 W$ and $\alpha^2 < 2$, the first term is at most $\beta W/128$. By (3), positive W is at least $1/k^2$, so the second term is also at most $\beta W/128$. Markov gives (17). □

The majority in (14) is taken after taking signs. Bounded output controls the exceptional set A even if the rational averages are large there. This argument does not assume an unbiased computed hypothesis or stability of an unbounded majority polynomial.

5 Complete algorithm and success probability

Set the following parameters, all determined by n, k:

$$\begin{aligned} T &= \lceil \log_2(2k) \rceil, & \beta &= \frac{1}{100(T+1)}, & R &= \lceil \sqrt{1/\beta} \rceil, \\ \varepsilon &= \frac{1}{512R}, & \gamma &= \frac{1}{32}, & m &= 144k, \\ \tau &= \frac{1}{4k}, & P &= 64k^2. \end{aligned} \tag{18}$$

Choose the least positive odd g such that $2^g \geq 512k^2/\beta$, and set

$$q_e = \left\lceil 32k\varepsilon^{-2} \ln \frac{4(k+1)}{\beta} \right\rceil,$$
$$q_c = \max\left\{ \left\lceil 32k\gamma^{-2} \ln \frac{4(P+1)}{\beta} \right\rceil, \left\lceil 300(k+1) \ln \frac{2(k+1)}{\beta} \right\rceil \right\}. \tag{19}$$

For an exact finite implementation, replace each $\ln z$ in a sample count by $\lceil \log_2 z \rceil$. This only increases the count and is computed by integer comparisons. The same convention applies to the search and final validation counts.

Initialize $U = \varnothing$ and $a = 0$. Every abort below returns FAIL. For exactly T rounds:

1. Measure q_e fresh labels. Store the residuals and v; use them to define (11).
2. Fourier-measure $4gm$ new copies. Discard every flag-zero outcome. If fewer than gm flags succeed, return FAIL. Split the first gm successful indices into g groups. Abort if more than k distinct indices occur among these gm grouped indices. Compute their estimates and construct (14). All estimates use the preceding label batch.
3. Apply Lemma 5 to the stored H, with τ, P, β above. If the search aborts, return FAIL; otherwise let L be its list.
4. Measure q_c new labels. Apply (7) to $L \setminus U$ with parameter γ, using residual $f - a$. Add all admissions permanently, including entries whose fitted coefficients may later be zero. Abort if $|U| > k$. Refit the target labels $f(X_\ell)$ from this same certification batch exactly on the updated archive U; abort on singularity or $\|a\|_2^2 > 2$.

After the last round, remove zero fitted coefficients, check sparsity at most k, and verify $a^2 = 1$ by exact Fourier convolution. The convolution has at most k^2 pairs; it does not enumerate the cube. Finally compare a with $\lceil 2k \ln 100 \rceil$ fresh labels. Return the exact list if all checks pass, and otherwise return FAIL.

Proposition 8 (Uniform contraction)**.** *Condition on any history with $U \subseteq S$ and (9). A round preserves these properties and satisfies $W(U^+) < W(U)/16$ when $W(U) > 0$, except on events of total probability at most 6β. For $W(U) = 0$, the archive retains S and the new fit remains exactly f with the same stated bound.*

Proof. There are six failure events: estimator accuracy, enough Fourier flags, the majority approximation, prefix search, original-coefficient certification, and the full-support Gram event. Their individual bounds are at most β, conditional on the good earlier stages where required. For flags, a binomial count with mean $2gm$ is below gm with probability at most $\exp(-gm/4) \leq \beta$: indeed, $gm/4 = 36kg \geq g \geq \log_2(1/\beta) \geq \ln(1/\beta)$. Conditional on enough flags the successful indices still have their independent Fourier law.

Suppose first $W > 0$. On the good search event, a true coefficient omitted from L has $|\widehat{H}(s)| < 1/(4k) \leq |c_s|/4$. Thus (17) implies

$$\sum_{s \in S \setminus L} c_s^2 \leq \frac{16}{9} \|H - f\|_2^2 \leq W/36. \tag{20}$$

For the proposed true coefficients still not admitted, Lemma 3 bounds each magnitude by $4\gamma e/\sqrt{k}$. Their total energy is at most $16\gamma^2 e^2 \leq \alpha^2 W/64$. Hence

$$W(U^+) \leq \left(\frac{1}{36} + \frac{\alpha^2}{64} \right) W < W/16. \tag{21}$$

No false frequency is admitted. The Gram event of Lemma 4 applies to the newly chosen archive using the same certification batch and restores (9). It also guarantees that the norm and singularity checks pass. There is no assumption that the new archive is independent of those rows.

If $W = 0$, the invariant gives $a = f$. The certification residual vanishes identically, so no new frequency is admitted. The Gram event makes the refit exact again; no majority-approximation assertion is needed. Fixed-round execution avoids making an inference about W from a sampled value $v = 0$. □

Proof of the success claim in Theorem 1. Initially $W = 1$ and (9) holds. A first-failure union bound over the T rounds gives failure probability at most $6T\beta < 0.06$. This conditions on good histories, without assuming independence of archives or round outcomes. On the complementary event,

$$W(U_T) < 16^{-T} \le \frac{1}{16k^4} < \frac{1}{k^2}.$$

Granularity forces $W(U_T) = 0$. The last exact refit is therefore f, and all final checks pass surely. This already gives success above 0.94, hence above 2/3.

For completeness, the final validation also limits erroneous acceptance. If two distinct Boolean functions have sparsity at most k, their difference d has at most $2k$ Fourier terms and $\|d\|_\infty = 2$. Cauchy–Schwarz gives $4 \le 2k\,\|d\|_2^2$, so their disagreement probability is at least $1/(2k)$. Conditional on any incorrect candidate passing Booleanity, the fresh validation accepts it with probability at most 1/100. Validation is not needed to increase the good-event success probability. □

6 All resource costs and exact arithmetic

The parameters satisfy $T, \beta^{-1}, \varepsilon^{-2}, g = O(\log(2k))$, $q_e = O(k\log^2(2k))$, and $q_c = O(k\log(2k))$. The total supplied-copy cap is

$$T(q_e + 4gm + q_c) + \lceil 2k \ln 100 \rceil = O(k\log^3(2k)). \tag{22}$$

The logarithm of n in the prefix-search confidence does not enter this count: those evaluations are of H. Each copy requires at most $O(n)$ single-qubit gates and measurements. No consumed copy is reused and no collective measurement is required.

Refits and coefficient precision. Every refit uses at most k columns of signs and the original sign labels. Its normal-equation Gram matrix and right side are integer, with entries bounded by q_c. Hadamard's determinant bound gives a common denominator and coefficient numerators of bit length

$$b_a = O(k\log(2kq_c))$$

on every nonsingular tape, regardless of whether the archive is correct. Fraction-free integer elimination computes the solution in polynomial bit time; singularity returns FAIL. Refitting resets this bit bound each round. No circuit for an earlier H is recursively embedded into the next one.

Estimator residuals share the fit's denominator. Therefore every subsequently evaluated estimate (11) has a common denominator and numerator length $O(b_a + \log(2kq_e))$. The variance estimate has a squared common denominator of comparable polynomial bit length. The cutoff, certification, norm and Booleanity tests use exact rational comparisons, with no numerical approximation.

Evaluating the hypothesis. At most k distinct grouped Fourier indices are stored; multiplicities specify their contributions to each group. The distinct-count cap never fires on valid ideal quantum examples, since each retained flag-one outcome belongs to S; discarded flag-zero outcomes are not counted. Write the nonzero estimates as N_s/D with a common positive integer denominator D. The product of their nonzero $|N_s|$ is a positive common denominator for the inverse weights, after incorporating D in the numerators. Its bit length is

$$O(k[b_a + \log(2kq_e)]) = \widetilde{O}(k^2).$$

Store the resulting signed integer weights and group multiplicities once. To evaluate H, compute the parities and signs of the group sums. Positive common factors and $1/m$ do not affect a sign. This costs $\widetilde{O}(nk + k^3)$ bit operations per query, using schoolbook integer arithmetic. The estimate cutoff prevents an uncontrolled statistical inverse weight; the common-denominator calculation also pays its exact computational representation cost.

Classical time, randomness and space. The search uses at most $2nP = O(nk^2)$ prefix tests. Each takes $O(k^4 \log(nk/\beta))$ samples. Consequently there are $O(nk^6 \log(nk/\beta))$ evaluations of H per round, which give (2) after charging the T rounds. Generating the shared-suffix input bits is included. Computing estimator residuals and requested coefficients, streaming candidate certificates, constructing the final at-most-k Gram matrix and exact elimination, and checking the final convolution have smaller polynomial costs than this conservative bound.

Store the current P prefixes or candidates, label rows for the estimator and certificate, the archive, sampled frequencies with multiplicities, integer weights, and streamed candidate accumulators. Candidate accumulators have $\widetilde{O}(k)$ bits each; there are $O(k^2)$ of them. The total is $\widetilde{O}(nk^2 + k^3)$ live bits. After admissions, compute the new Gram matrix from the stored certificate rows; a P-by-P matrix is unnecessary. All temporary integer arithmetic fits these bounds. Old hypotheses and batches are discarded once the new fit is available.

Every round count, batch size, prefix count and arithmetic input size is capped. On bad histories the known function H is still Boolean, since it is a sign-majority. Zero inverses, ties, singular systems and archive overflow have explicit rules. Thus every tape terminates within the stated polynomial bounds, not merely every successful tape. This completes Theorem 1.

7 Scope, limitations and verification status

The algorithm bounds both sample use and computation in the noiseless model of supplied examples. It does not improve Ye's logarithmic sample factor [2], achieve the $O(k \log k)$ target, or provide a practical implementation. Its constants and polynomial exponents are deliberately loose. Noise robustness and improved classical runtime require separate analysis.

The known H can have degree n and exponentially many Fourier coefficients. This is permitted because its evaluation cost is polynomial and it only proposes indices. The sparse residual used for estimation and certification is always $f - a$. The proof does not require a generic stability theorem for nonlinear corrections of sparse fits, or a short representation of every recovered index in the original samples.

After measurement, the algorithm is classical: its data consist of uniform labeled examples and successful Fourier samples with law c_s^2. The access model does not yield a general classical simulation of quantum computation or decide BPP versus BQP. Quantum-example preparation is not included in the theorem. Comparison with classical random-example bounds must keep this difference in supplied data explicit.

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